

Performance evaluation of sensors for gaseous pollutants and particulate matter in ambient air: *European standardization and current work at EU level*

Martine Van Poppel

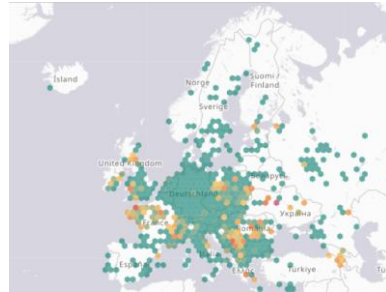
VITO

Sensor systems

sensors - sensor systems - sensor solutions - hybrid systems - ...



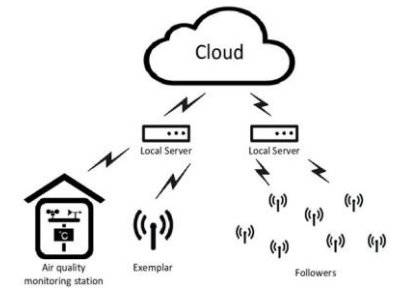
Sensors
Different types
Measuring unit



DIY building blocks
sensor, Arduino,
dataportal



Sensor systems
Sensor solutions
Sensor unit, database and dataportal,
calibration factors
visualisation



Hybrid systems
Sensor network
using also additional data
(a.o. AQMN, models)

EU Air Quality Directive

- EU Air Quality Directive: 2008/50/EC
 - Objectives for AQ limit values and **common methods** to assess AQ
 - **PM₁₀, PM_{2,5}, NO₂, CO, O₃, SO₂, Pb, Benzene**
 - It includes Data Quality Objectives for measurement techniques:
 - Reference monitoring (fixed sampling points)
 - **Indicative measurements**
 - **Objective estimations**
- **To supplement fixed sampling points**
Sensors can play a role here (but NOT ONLY HERE)



CEN standardization work on sensor evaluation

- Developed by TC264/**WG42** (Ambient Air - Air Quality Sensors)

- Aim:
 - Standardized approach for sensor system **testing and classification**
 - Provide a **test protocol** and **classification scheme**

- Applies to
 - Sensor systems as **individual** measurement devices for **outdoor** measurements at **fixed** sites
 - Sensor systems as **indicative measurements** or **objective estimations** (related to EU AQD)

Status of standardization work

- Air quality — Performance evaluation of air quality sensor systems —
- Part 1 Gaseous pollutants in ambient air (NO₂, NO, CO, SO, O₃, benzene, CO₂)
 - CEN/TS 17660-1 available
- Part 2 Particulate Matter in ambient air
 - CEN/TS 17660-2 available
- Validation work: TS → CEN Standard : on-going



Data Quality Objectives sensor systems

In CEN/TS 17660-1&2

DQO	O ₃	CO, NO ₂ , SO ₂	PM ₁₀ , PM _{2.5}
DQO Reference measurements	U = 15%	U = 15%	U = 25%
DQO Indicative measurements	U = 30%	U = 25%	U = 50%
DQO Objective estimations	U = 75%	U = 75%	U = 100%
	U = 200%	U = 200%	U = 200%

SENSOR CLASSES

————→ CLASS 1

————→ CLASS 2

————→ CLASS 3

Directive 2008/EC/50

→ some changes in New AQD!

U: expanded uncertainty

Averaging time: period considered by the Limit Value (LV)

-> 1h (gas)

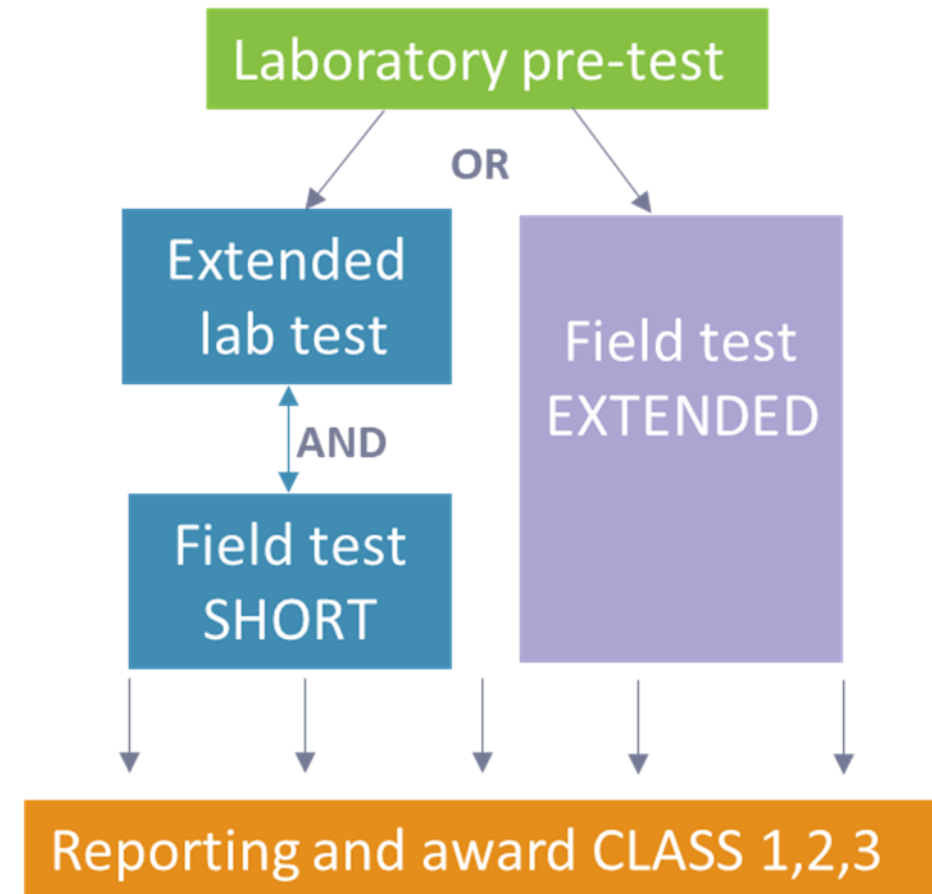
-> 24h for PM!

CEN/TS 17660-1

Gas sensors

Overview evaluation protocol (gas)

- Laboratory pre-test:
 - Response time, lack of fit, LOD, repeatability
- Extended lab tests
 - long-term drift
 - Cross-sensitivity to other pollutants
 - T and RH interference
- Field test
 - Between sensor uncertainty
 - Data capture
 - expanded uncertainty of field tests => DQO at LV
 - Slope, intercept of regression line
- Sensor systems tested: 3 replicates



Laboratory tests

- Protocol defines
 - Gas concentration levels (4-5)
 - Interferences to be tested
 - Tested at zero and span
 - Levels of T, RH :
 - T: 4 levels between -20 – 40°C
 - RH: 10-25%; 40-50%; 70-75%; 90%
- Expanded uncertainty of all lab uncertainties



Field tests (gas)

- Protocol defines
 - Number and type of test sites per pollutant
 - Test conditions:
 - Meteorological conditions: 2 seasons (min 40 days)
 - Concentration levels ($> 50\%$ of 5y-average P98)
 - Installation, on-going quality control
 - Correction for slope and intercept



Calculation of uncertainties - field

- Between sensor uncertainty (field)
 - Determined per field site
 - For both seasons together
- Expanded uncertainty of field tests at Limit value L:
 - Dataset at different AQMS are evaluated individually
 - Sensor systems are evaluated individually
 - Only a single slope and intercept correction are allowed;
(for all systems and sites)

$$u(bs, s) = \sqrt{\left(\frac{\sum_{i=1}^n \sum_{j=1}^{n_i} (y_{i,j} - \bar{y}_i)^2}{n(n_s - 1)} \right)}$$

$$U_{\text{field},L} = k \times \sqrt{\underbrace{\left(\frac{R}{n-2} - u(bs, RM)^2 \right)}_{\text{RANDOM}} + \underbrace{\left[a + (b-1) \times L \right]^2}_{\text{BIAS at limit value (L)}}}$$

RANDOM
Residuals of regression line
and correction for between instrument uncertainty
of reference method

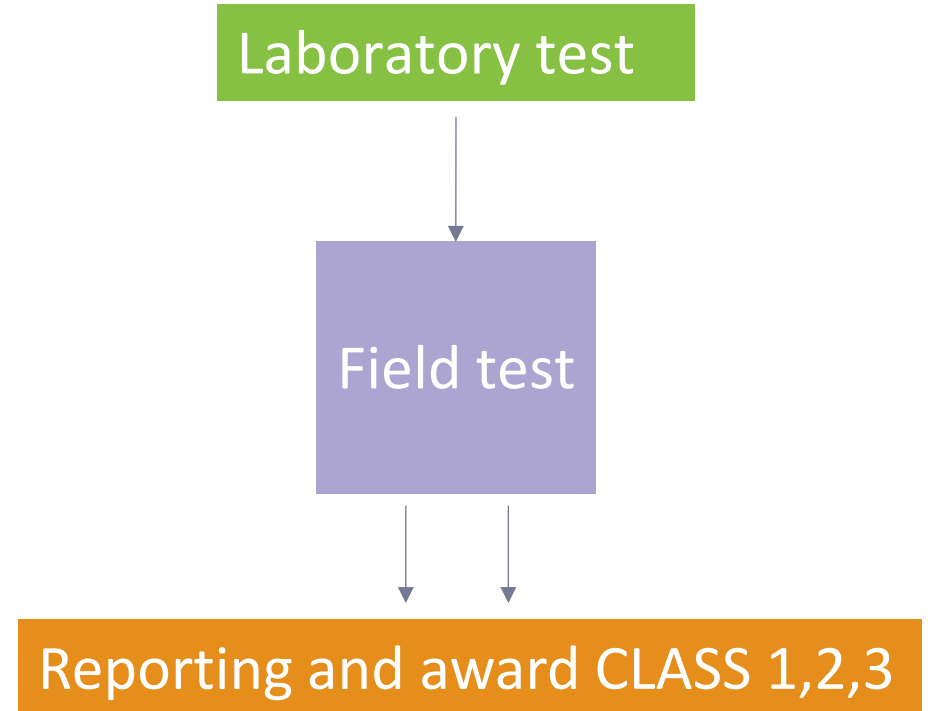
BIAS at limit value (L)
Bias between regression line
and perfect agreement
(b=1 and a=0)

CEN/TS 17660-2

PM sensors

Overview evaluation protocol (PM_{2,5} and PM₁₀)

- Main differences with gas:
 - Reference method is filter based (24h)
 - PM ≠ gas
- Focus on field tests
 - Laboratory test: PM coarse
- Performance requirements:
 - Between sensor uncertainty (BSU)
 - Data capture
 - expanded uncertainty of field tests => DQO at LV (**24h**)
 - **PM coarse (lab and field)**
 - **RH**
 - Slope, intercept of regression line
- Sensor systems tested: 3 replicates



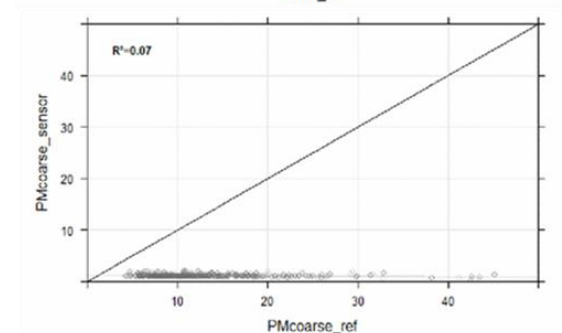
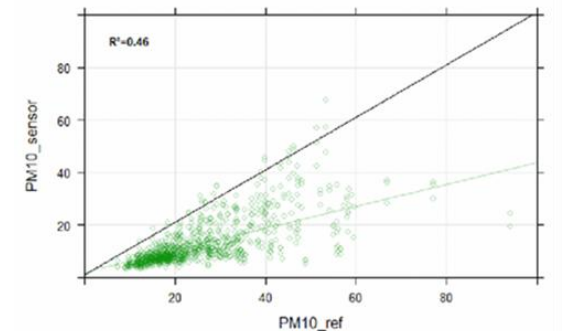
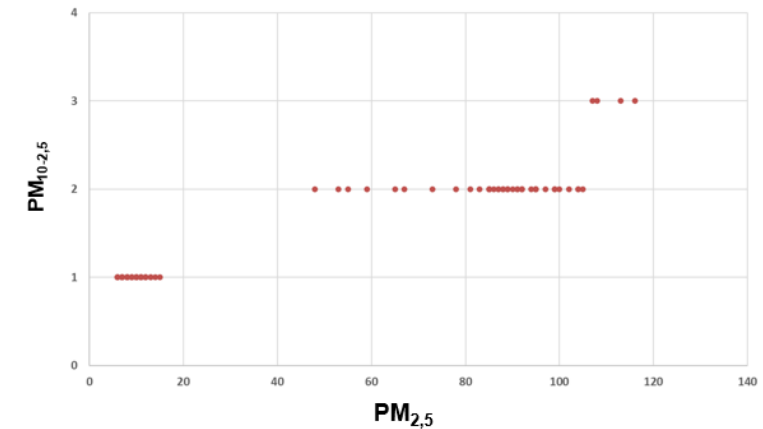
Field tests (PM)

- Protocol defines
 - Number and type of test sites (**2**): **traffic, background**
 - Test conditions:
 - Meteorological conditions: 2 seasons (min 40 days)
 - Concentration levels (> 50% of 5y-average P98)
 - **PM₁₀: at least one site with significant PM_{10-2,5}**
 - **Reference method (24h) and equivalent method (1h)**
 - Installation, on-going quality control
 - Correction for slope and intercept



PM coarse problem

- $PM_{coarse} = PM_{10} - PM_{2,5} = PM_{10-2,5}$
- Some sensors do not measure PM_{10} but **calculate PM_{10}** from measured $PM_{2,5}$
- **PROBLEM:**
 - At most locations where sensors are tested, PM_{coarse} is relatively low ($PM_{10} \sim PM_{2,5}$) and PM_{10} values will be acceptable
 - These sensors do **not** pick up PM_{10} peak concentrations at occasions where $PM_{10} \gg \gg \gg PM_{2,5}$
- **SOLUTION:**
 - Simulate high PM_{coarse} conditions in the lab
 - Compare sensor/reference ratio at high and low PM_{coarse} conditions



PM coarse test (lab) - setup

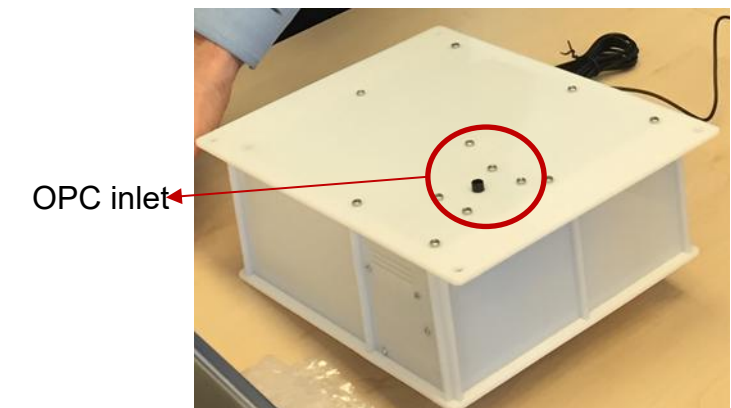
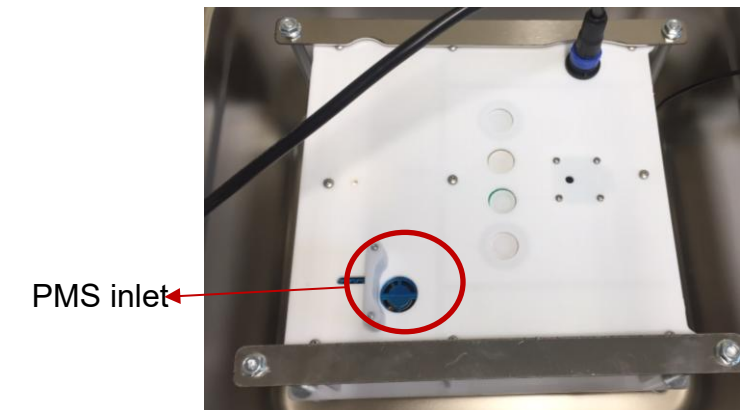
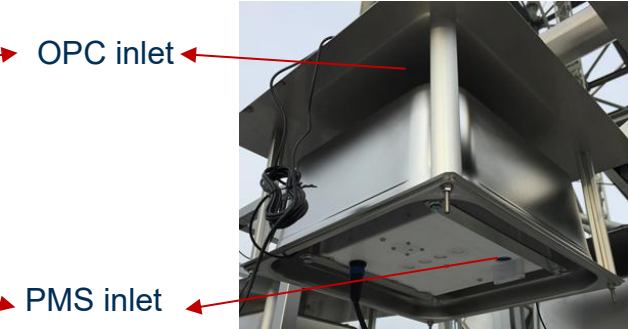
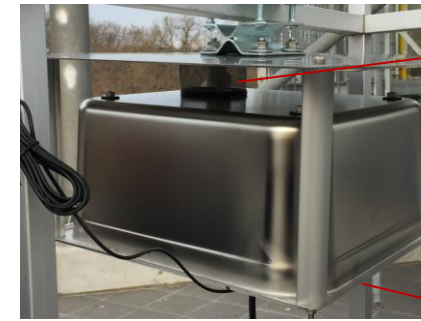
- Two tests with different particle size
 - **Coarse test:** $>80\%$ $PM_{10-2,5}$ in PM_{10}
 - **Fine test:** $< 20\%$ $PM_{10-2,5}$ in PM_{10}
- Sensor system has to show a strong response to different fractions
 - ⇒ Sensor response is compared to equivalent method for both tests
 - ⇒ Change in sensor response from fine to coarse conditions is calculated



Example: AirsensEUR boxes

- AirSensEUR version 3.0.
- AirSensEUR is an Open Platform project developed by the JRC, aimed at measuring air quality using low-cost sensors
- AirSensEUR includes a PTFE enclosure (26x22x10cm), inserted in a stainless-steel protecting cover (35x32x30cm)
- PM sensors:
 - **OPC-N3** with 24 size bins (0.3/0.35 – 40 μm) with a counting efficiency* of 50% @ 0.3 μm and 100% @ 0.35 μm
 - **PMS5003** with 6 size bins (> 0.3 μm) with a counting efficiency* of 50% @ 0.3 μm and 100% @ 0.5 μm

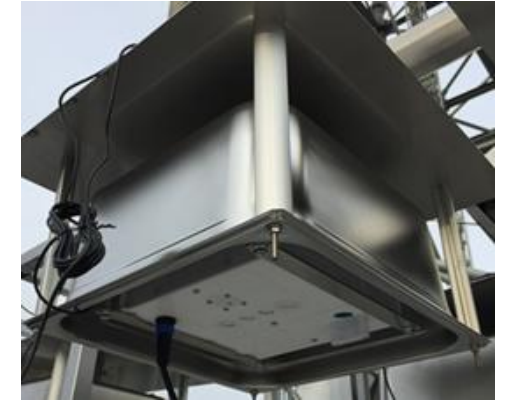
*counting efficiencies claimed by the manufacturers



Example: two different PM sensors



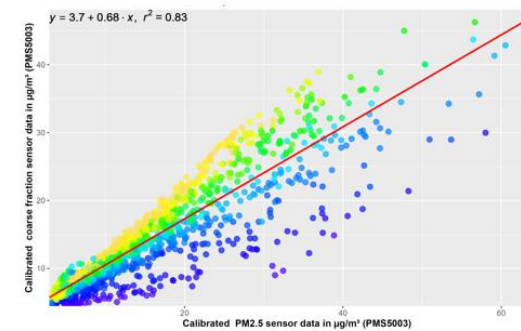
$R_{PM_{2,5}(FINE)} = \frac{PM_{2,5}(sen,FINE)}{PM_{2,5}(eq,FINE)}$	→	0.27	0.92
$R_{PM_{10-2,5}(COARSE)} = \frac{PM_{10-2,5}(sen,COARSE)}{PM_{10-2,5}(eq,COARSE)}$	→	0.06	1.42
$F\left(\frac{FINE}{COARSE}\right) = \frac{R_{PM_{2,5}(FINE)}}{R_{PM_{10-2,5}(COARSE)}}$ < 2 for Class 1!	→	4.3 >2 FAIL	0.65 <2 PASS



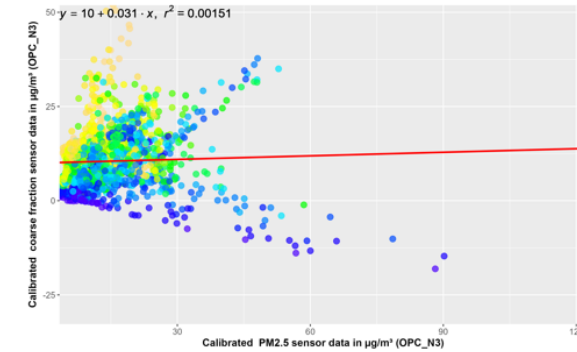
AirsensEUR boxes: field data

- $PM_{10-2,5}$ versus $PM_{2,5}$
 - No linear relationship observed for equivalent (reference): $PM_{10-2,5}$ versus $PM_{2,5}$
 - PMS5003 shows linear relationship
- PMS sensor versus equivalent (reference):
 - Clear linear relationship for $PM_{2,5}$ no linear relationship for $PM_{10-2,5}$
- Observations of lab tests are confirmed

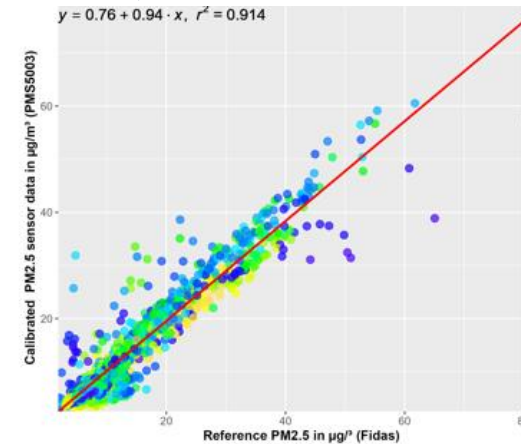
$PM_{10-2,5}$ vs $PM_{2,5}$: PMS



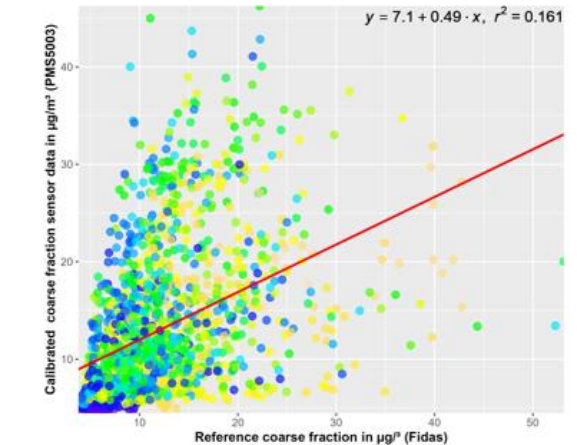
$PM_{10-2,5}$ vs $PM_{2,5}$: OPC



$PM_{2,5}$: PMS vs reference

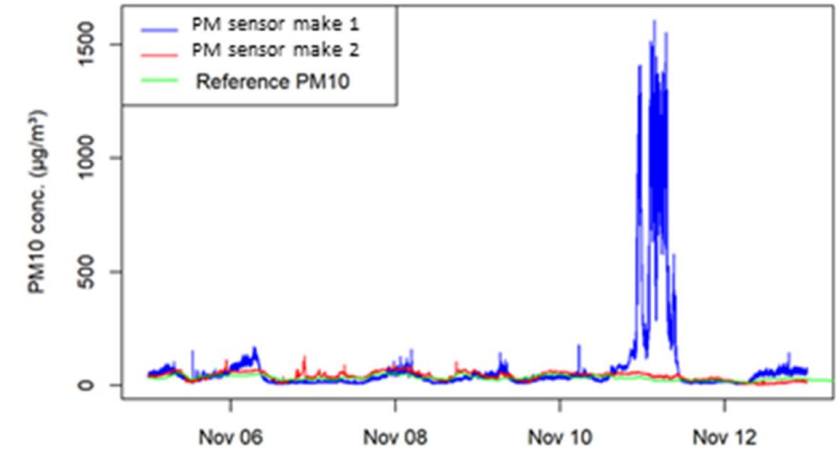


$PM_{10-2,5}$: PMS vs reference



RH problem

- Some sensors overestimate PM readings at high RH
- **PROBLEM:**
 - When only looking at 24h values 'over' readings can be compensated by 'under' readings during other parts of the day
 - No RH issue will be observed based on 24h averages whereas strong overestimations can occur at hourly values
- **SOLUTION:**
 - Plot hourly average sensor readings as function of RH
 - Compare sensor/reference ratio at 'dry' and 'humid' conditions in the field

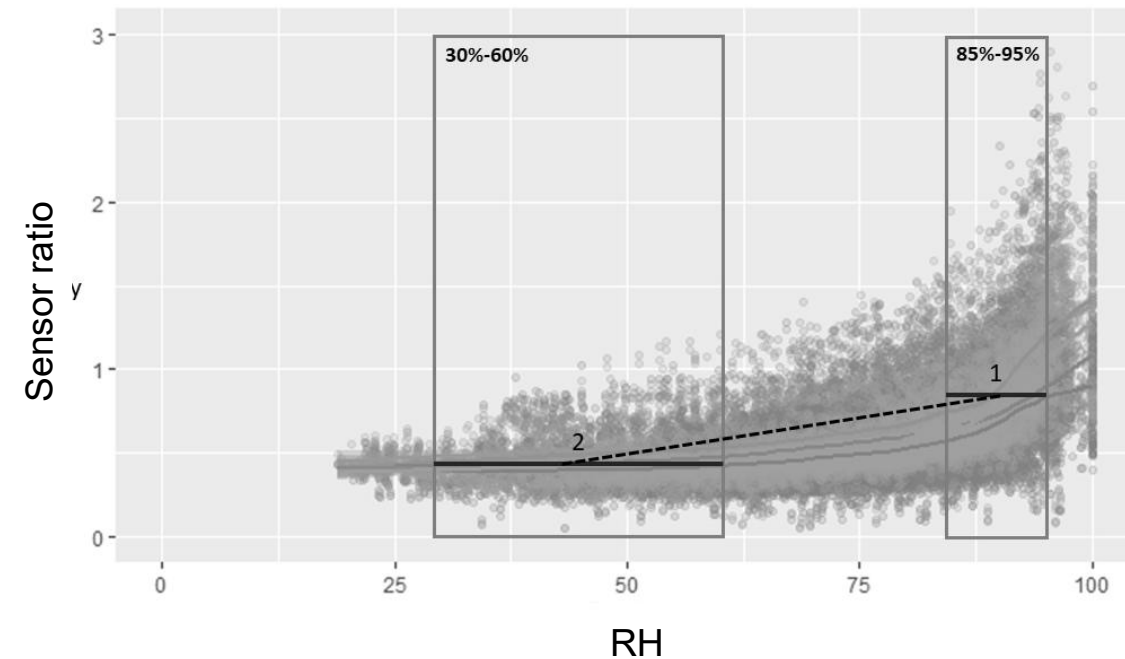


RH effect (calculations)

- Field data
- Calculate sensor ratio:
 - ratio of sensor value to equivalent value
 - using hourly values
- RH factor is defined as change in sensor ratio from low (30-60%) to high (85-95%) RH

$$RH_{factor} = \frac{Mdn(sensorratio)_{85-95\%}}{Mdn(sensorratio)_{30-60\%}}$$

$$Sensor\ ratio = \frac{y_{sen}}{y_{eq}}$$



Class 1: $RH_{factor} < 1.5$

Class 2: $1.5 < RH_{factor} < 2$

Classification of sensor systems

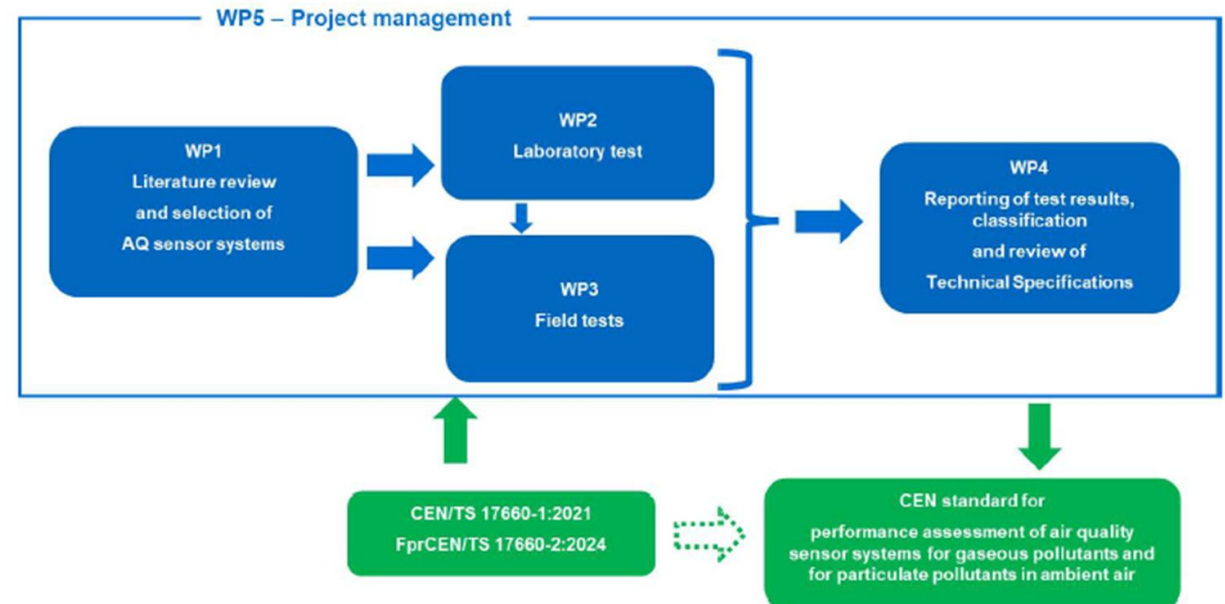
- Classification is based on **DQO** and **performance requirements**
 - DQO (as maximum expanded uncertainty):
 - **Class 1**: DQO of indicative measurements
 - **Class 2**: DQO of objective estimations
 - **Class 3**: more relaxed (200%), not linked to Directive, e.g. for research, education, ...
 - Performance requirements:
 - response time, lack of fit, repeatability, LOD
 - between sensor uncertainty, data capture
 - coarse test, RH test (PM)
- => More stringent for **Class1 > Class 2 > Class 3**



Validation work

- Project DGENv: “Support for the development of performance assessment of air quality sensor systems for gaseous pollutants and particulate pollutants in ambient air” (EC-ENV/2024/OP/0047)
 - 2 sensor systems tested according to TS
 - In two different laboratories (lab tests)
 - Different meteorological conditions (field tests)

City	Site name	Area type	Site type
Oslo	Kirkeveien	urban	traffic
	Chilbolton	rural	background
Gif-sur-Yvette	SIRTA	suburban	background
Rome	Villa Ada	urban	background
	Fermi	urban	traffic
	Allumiere	rural	background



Summary and outlook

- CEN TS 17660-1/2 (test protocol and classification) is available for gas and PM
- Differences between test protocols for gases and PM
 - TS for PM sensors mainly focused on field evaluation
 - Lab testing included for coarse test (PM)
- Sensor evaluation is done on individual sensor systems (3) and tests
 - The lowest classification is given as general classification
- Current TS is focused on
 - use of sensors in regulatory context and individual sensors (not networks)
- Future work:
 - on-going QA/QC -> current work of WG42
 - Other: sensor in sensor networks (calibration), non-regulated pollutants, mobile sensors,...
 - Alignment with new AQD

QUESTIONS?



Contact: martine.vanpoppel@vito.be